Green Campus Annual Progress Report 2024: Key Performance Indicators

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GREEN CAMPUS ANNUAL PROGRESS REPORT 2024: KEY PERFORMANCE INDICATORS

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1. EXECUTIVE OVERVIEW AND 2024 OPERATIONAL SCORECARD

Vivekananda Education Society's Institute of Technology (VESIT Green Campus) in Chembur, Mumbai, concludes calendar year 2024 having established comprehensive baseline carbon, energy, water, and waste accounting across its 32,500 m2 built-up infrastructure. Serving a vibrant community of 3,200 individuals (3,000 undergraduate/postgraduate students and 200 faculty/staff), the institution achieved major milestones in evidence-grounded sustainability.

2024 Core Institutional Performance Metrics:

- Total Institutional Carbon Footprint: 2,450,000 kg CO2e (2,450.00 metric tonnes CO2e).

- Scope 1 Direct Emissions: 49,312 kg CO2e (2.01%), driven by 18,400 L of diesel consumed in emergency generators during regional grid dropouts (EF_DIESEL_GEN = 2.68 kg CO2e/L).

- Scope 2 Indirect Grid Electricity Emissions: 1,681,000 kg CO2e (68.61%), from importing 2,050,000 kWh from the Maharashtra grid (EF_ELEC_GRID = 0.82 kg CO2e/kWh).

- Scope 3 Value Chain Emissions: 719,688 kg CO2e (29.38%), comprising commuting (580,000 kg CO2e), municipal landfill waste (115,000 kg CO2e), and water pumping (24,688 kg CO2e).

- Total Electricity Consumed: 2,362,400 kWh (Grid: 2,050,000 kWh | Solar PV: 312,400 kWh).

- Clean Renewable Electricity Share: 13.22% of total campus electricity demand satisfied by on-site rooftop solar.

- Gross Freshwater Intake: 48,000,000 litres drawn from municipal supply.

- Rainwater Harvested and Reused: 12,400,000 litres captured across terrace roofs, satisfying 25.83% of annual campus water needs.

- Total Solid Waste Generated: 68,400 kg.

- Solid Waste Diversion Rate: 48.2% diverted from landfills via on-site composting (19,200 kg) and dry recycling (19,800 kg).

- Low-Carbon Commuter Share: 52.0% of trips delivered by train, municipal bus, EV shuttle, and cycling.

2. DETAILED PROGRESS AGAINST STATUTORY SUSTAINABILITY TARGETS

- Target Net-Zero (TRG_NET_ZERO): Decarbonize baseline 2,450,000 kg CO2e to net-zero by 2040. In Progress. The linear abatement trajectory mandates eliminating 153,125 kg CO2e annually. First-year initiatives achieved 256,168 kg CO2e of avoided solar emissions and 36,000 kg CO2e of transport abatement, exceeding year-one milestones.

- Target Renewable Electricity (TRG_RENEWABLE): Attain 50.0% clean power by 2030. In Progress. Phase-1 Solar (INIT_SOLAR_PH1, 220 kWp) generates 312,400 kWh/year. Board clearance for Phase-2 Solar (INIT_SOLAR_PH2, 200 kWp + 100 kWh BESS) will escalate clean energy generation to 596,400 kWh/year (25.25% penetration).

- Target Zero Waste to Landfill (TRG_ZERO_WASTE): Achieve 85.0% landfill diversion by 2028. In Progress. 48.2% diversion achieved in 2024 via Canteen biogas digester (INIT_BIOGAS_CANTEEN) and paper recycling.

- Target Water Neutrality (TRG_WATER_NEUTRAL): Reduce municipal water intake by 30.0% by 2029. In Progress. 12.4 million litres harvested via INIT_RWH_SYSTEM; plumbing aerator retrofits initiated across 140 washrooms.

- Target Green Commuting (TRG_GREEN_COMMUTE): Achieve 70.0% low-carbon commuting by 2027. In Progress. Two Mahindra Supro electric vans (INIT_ELECTRIC_SHUTTLE) delivered 38,400 clean passenger-trips.

3. RESEARCH INNOVATION: THE ECOCLOCK AI PLATFORM

A primary institutional highlight was the joint development and deployment of the EcoClock AI platform by the Department of Computer Engineering (DEP_CMPN) and Department of AI & Data Science (DEP_AIDS). EcoClock AI integrates persistent Neo4j knowledge graphs, IoT telemetry from 45 Modbus smart meters (INIT_IOT_ENERGY_MGT), and local Mistral LLM reasoning to deliver complete transparency, interactive scenario simulation, and evidence-grounded verification. The Governing Board commends all academic departments, faculty, and student ambassadors for their pioneering environmental leadership.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_ADM_020**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*